

LESSON 8.5

SOLVING REAL-WORLD EQUATIONS WITH VARIABLES ON BOTH SIDES



LESSON INTRODUCTION

MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

LEARNING TARGETS

- I can solve equations with variables on both sides in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

WORKING TOWARD

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

ESSENTIAL QUESTIONS

- How can equations with a variable on both sides be used to represent real-world problems?
- How can you use equations to solve real-world problems?

LESSON NARRATIVE

This lesson focuses on real-world contexts. Students continue to practice solving equations with variables on both sides, but in this lesson, each equation represents a real-world context. Students will explore how to relate an equation to a real-world context and use the solution to solve problems. This lesson presents opportunities for students to make connections between real-world contexts and mathematical expressions and equations as they build a stronger understanding of the utility of equations.

COMPONENT SUMMARY

8.5.1 WARM-UP (~5 MIN.)

Reflect on how mistakes can be learning opportunities

8.5.2 EXPLORATION (15 MIN.)

Explore how to use an equation to solve a real-world problem

8.5.3 GUIDED INSTRUCTION (10-15 MIN.)

Solve equations representing real-world contexts

8.5.4 GUIDED PRACTICE (5-10 MIN.)

Practice solving equations representing real-world contexts

8.5.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of solving equations representing real-world contexts

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- (Optional) Highlighters or colored pencils

ADDITIONAL PREPARATION

- Warm-Up 8.5.1 contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may try to combine like terms across the equal sign.
- Students may struggle to connect the given equations to the situations they represent.
- Students may struggle to understand the meaning of the solution in context.

THINGS TO CONSIDER

- Model how to make connections between the given equations and the situations they are modeling.
- Provide students with opportunities to discuss these connections.
- Consider ways to spiral connections to linear relationships from Unit 7 throughout this lesson. While students do not need to write the equations themselves at this point, building these connections will help them to make sense of how to do so in Algebra 1.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

The Exploration invites students to use an equation to solve a problem involving water tanks. Consider applying Notice and Wonder routines to the problem, encouraging students to discover the relationship between the information presented in the table and the representation and solution of it in the equation.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students will be working to connect equations to the real-world situations they are being used to represent. Clarifications within this standard emphasize using models and methods to represent everyday experiences. Utilize this lesson to provide opportunities for students to use equations with variables on both sides to represent and solve real-world problems.

DIFFERENTIATE INSTRUCTION

Allow students to use highlighters or colored pencils to make connections and annotations between the situations and the equations used to model them.

In the 8.5.2 Exploration, consider having students model the water in the tanks by moving across the classroom floor. Have one student represent Tank 1 by slowly moving downward across the classroom, while the student playing Tank 2 moves upward across the classroom so that students are able to see that at a certain point, they will be in the same place.

8.5.1 WARM-UP: MISTAKES ARE LEARNING OPPORTUNITIES

TEACHER MOVES

If time allows, consider pairing students to discuss their reflections.



8.5.1 Warm-Up: Mistakes are Learning Opportunities

1. Describe a time in your life when you made a mistake, but it was a good thing because you were able to learn from that mistake.

Answers will vary.

2. How does making mistakes when solving math problems help you?

Answers will vary.



8.5.2 Exploration: Water Tanks

Work with your partner to complete the following.

- The amount of water in two tanks, measured every 5 minutes, was recorded in the table shown.

Time (minutes)	Tank 1 (liters)	Tank 2 (liters)
0	25	1000
5	175	900
10	325	800
15	475	700
20	625	600
25	775	500
30	925	400
35	1075	300
40	1225	200
45	1375	100
50	1525	0

- Review the table with your partner and describe the changes in water levels for each tank over time.

Tank 1	<i>Answers will vary. The water level in tank 1 is increasing 150 liters every 5 minutes. The water level in tank 1 is increasing by 30 liters per minute.</i>
Tank 2	<i>Answers will vary. The water level in tank 2 is decreasing 100 liters every 5 minutes. The water level in tank 2 is decreasing by 20 liters per minute.</i>

- Based on the table, predict the approximate number of minutes after which the tanks will reach the same water level.

Answers will vary. It seems like they will be at the same level somewhere between 15-20.

- Estimate the amount of water in each tank when their levels are the same.

Answers will vary. There will be a little more than 600 liters of water in the tank when they are at the same level.

- Since one tank is leaking water and the other is filling with water, at some point they will have the same amount of water in each. The water level in Tank 1, after t minutes, can be represented by the expression $30t + 25$. The water level in Tank 2, after t minutes, can be represented by the expression $-20t + 1000$. The equation $30t + 25 = -20t + 1000$ can be used to determine when the water levels are the same.

- Complete the missing steps to determine after how many minutes, t , the water levels will be the same in each tank.

$30t + 25 = -20t + 1000$
$50t + 25 = 1000$
$50t = 975$
$t = 19.5$

- Use the solution to complete the missing steps to determine how many liters of water were in each tank when their levels were the same.

Tank 1	Tank 2
$30t + 25$	$-20t + 1000$
$30(19.5) + 25$	$-20(19.5) + 1000$
$585 + 25$	$-390 + 1000$
610 liters	610 liters

- Compare your estimates to the actual calculated times and water levels when the tanks contained the same amount of water.

Which estimates were the closest, yours or your partner's?

Answers will vary.

8.5.2 EXPLORATION: WATER TANKS

TEACHER MOVES

8.5.2 Exploration is scaffolded to allow students to engage with their partner rather than being guided through by the teacher. They are positioned to first consider the problem in context before being presented with an equation to solve.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share whole-group in 8.5.3 Guided Instruction. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

ENRICHMENT

For students who are stuck, consider asking the following:

- After how many minutes are the water levels in the tanks most similar?
- Based on the volumes of the tank before and after that time, do you think they would have been at the same level before or after that time?

TEACHER MOVES

Consider pausing to have students relate the expressions to the linear relationships in the table if time allows. This provides an excellent opportunity to spiral content from Unit 7 relating the rate of change and initial value of the volume of water in each tank to the expression used to model it.

QUESTIONING

Challenge students to prove their algebraic solutions graphically by allowing them to use a graphing utility. Students should graph both lines and determine where the water levels are equivalent by identifying the point of intersection.

8.5.3 GUIDED INSTRUCTION: SOLVING REAL-WORLD EQUATIONS WITH VARIABLES ON BOTH SIDES

TEACHER MOVES

Consider giving students an opportunity to read the problem on their own first and discuss the situation with a partner. Then, explicitly make the connection between the expressions and the contexts by using highlighters or other annotations of the text. For question 1a, ask students to consider why the two expressions were set equal to each other and what that means in this context. After giving students a chance to discuss their ideas, have them read question 1a and solve the equation.

Model how to interpret the solution in context and use the solution to determine the final cost for the trophy at either store.



8.5.3 Guided Instruction: Solving Real-World Equations with Variables on Both Sides

- At Winning Trophies, a trophy costs \$12.50 and engraving costs \$0.40 per letter. At Sport Awards, the same trophy costs \$14.75 and engraving costs \$0.25 per letter.

The situation can be represented by the following expressions, where e is the number of letters engraved.

Winning Trophies: $12.50 + 0.40e$

Sport Awards: $14.75 + 0.25e$

- Solve the equation $12.50 + 0.40e = 14.75 + 0.25e$ to find how many letters need to be engraved on the trophies for the final cost to be the same at both stores.

$$\begin{aligned} 12.50 + 0.40e &= 14.75 + 0.25e \\ 0.40e &= 2.25 + 0.25e \\ 0.15e &= 2.25 \\ e &= 15 \end{aligned}$$

- Check your work by determining the final cost for the trophy at either store.

Cost at Winning Trophies	Cost at Sport Awards
$12.50 + 0.40e$	$14.75 + 0.25e$
$12.50 + 0.40(15) = \$18.50$	$14.75 + 0.25(15) = \$18.50$

2. Keely is hiking at an altitude of 13,000 feet and is ascending 20 feet per minute. Leonard is hiking the same trail but is at an altitude of 13,750 feet and is descending 30 feet each minute.

If m represents the number of minutes, the equation $13000 + 20m = 13750 - 30m$ can be used to determine when the hikers will meet.

- a. Solve the equation $13000 + 20m = 13750 - 30m$.

$$13000 + 20m = 13750 - 30m$$

$$50m = 750$$

$$m = 15$$

- b. In terms of the situation, what does the value of m represent?

The value of m represents the number of minutes that have passed when the two hikers meet.

- c. At what altitude, in feet, will the two hikers meet?

$$13000 + 20(15)$$

$$13000 + 300 = 13,300$$

The two hikers will meet at an altitude of 13,300 feet.

8.5.3 GUIDED INSTRUCTION: SOLVING REAL-WORLD EQUATIONS WITH VARIABLES ON BOTH SIDES (CONTINUED)

TEACHER MOVES

Students may struggle to understand how the equation represents when the hikers will meet. Have students locate the language showing that the hikers will meet on the trail with each person coming from a different direction.

Consider allowing students to highlight or make annotations to note connections between the text and the expressions set equal to each other. Observe as they make their markings to see which connections students are making and which may need additional clarification before completing questions 2a-c.

8.5.4 GUIDED PRACTICE: SOLVING REAL-WORLD EQUATIONS WITH VARIABLES ON BOTH SIDES

TEACHER MOVES

Consider a variety of flexible groupings for the activity. If students are able to complete the activity independently, allow them to do so. For students who need support, place them with a partner or in a group of three. As students work, circulate throughout the class to observe how comfortable students feel solving the equation, interpreting the solution, and using the solution to solve question 1b.

ENRICHMENT

Consider asking the following questions to support students making connections between the context and the equation:

- Whose plan does $35 + 10m$ represent?
- Whose plan does $13.50m$ represent?
- Why are the two expressions set equal to each other in this case? How does that relate to the meaning of the variable?
- How can you use the solution to the equation to complete question 1b?



8.5.4 Guided Practice: Solving Real-World Equations with Variables on Both Sides

- Stephanie has a VIP membership to a movie theater, so she pays \$35.00 a year and \$10.00 for each movie she sees. Jeff doesn't have a membership, so he pays \$13.50 for each movie he sees.

- Solve the equation $35 + 10m = 13.50m$ to determine how many movies, m , they each must see in a year for their costs to be the same.

$$35 + 10m = 13.50m$$

$$35 = 3.50m$$

$$10 = m$$

They spend the same amount of money after seeing 10 movies.

- Use the solution to find how much money Stephanie and Jeff will each spend to see this number of movies.

Stephanie and Jeff will each spend a total of \$135 dollars to see 10 movies.



8.5.5 Wrap-Up: Ticket Out the Door

1. The cost for renting movies from two different online streaming platforms is shown in the table.

Movies Rented	Video Prime (\$)	Movie Flix (\$)
0	50.00	0
1	53.00	5.50
2	56.00	11.00
3	59.00	16.50
4	62.00	22.00
5	65.00	27.50

Video Prime charges a \$50.00 membership fee, plus an additional \$3.00 charge per rental. Movie Flix charges no membership fee, but charges \$5.50 per rental.

- a. Use the equation $50 + 3m = 5.50m$ to determine how many movies need to be rented for the cost to be the same, where m represents the number of movies rented.

$$50 + 3m = 5.50m$$

$$50 = 2.50m$$

$$20 = m$$

After renting 20 movies, Video Prime and Movie Flix will cost the same.

- b. How much will a customer pay when the cost for each platform is the same?

$$5.50(20) = 110 \text{ or } 50 + 3(20) = 110$$

The customers will pay \$110 for 20 movies.

8.5.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

As students are working, notice what aspects of the problem students feel comfortable working with and where students hesitate or struggle. Use students' responses to guide future instruction.